Probability of Default (PD) Modeling Using Machine Learning

Capstone Project

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1. Introduction

Financial institutions issue thousands of consumer loans every year, making effective credit risk assessment essential for maintaining a healthy lending portfolio. Although loans are approved using underwriting criteria, some borrowers will inevitably experience financial hardship and default on their obligations. Accurately estimating the probability of default enables lenders to understand portfolio risk better, allocate capital more effectively, and make more informed risk management decisions.

For this project, I used the historical LendingClub loan data from Kaggle to develop and compare several machine learning classification models for estimating the probability of loan default. Rather than evaluating new loan applications, the objective is to assess the risk of an existing portfolio by assigning each borrower a probability of default.

Multiple machine learning algorithms—including Logistic Regression, Random Forest, and XGBoost—were developed and compared using metrics appropriate for highly imbalanced classification problems. The final model was selected based on its ability to balance predictive performance with practical business considerations.

2. Project Objective

The objective of this project was to develop and evaluate machine learning models to estimate the probability of default for issued loans. The analysis uses borrower characteristics, credit history, and loan-specific information to identify patterns associated with subsequent default risk.

To ensure that model performance reflects predictive relationships rather than information that directly reveals the loan outcome, variables associated with realized payment performance, recoveries, collections, and other post-outcome information were excluded from model development. This reduces the risk of target leakage and provides a more realistic assessment of the relationship between borrower and loan characteristics and probability of default.

Several classification algorithms were trained, tuned, and compared using metrics appropriate for an imbalanced credit-risk problem. Particular emphasis was placed on identifying defaulted loans while maintaining reasonable classification performance for non-defaulted loans and overall discriminatory ability.

3. Business Context

Credit risk is one of the primary risks faced by banks and other lending institutions. While every loan generates interest income, borrower defaults can significantly reduce portfolio profitability and increase capital requirements.
Predictive models that estimate the probability of default help financial institutions monitor the health of existing loan portfolios, identify higher-risk borrowers, and support risk management decisions. Rather than replacing traditional credit analysis, these models provide an additional quantitative tool for evaluating portfolio performance.

4. Intended Audience

This project is intended for students and professionals interested in applying machine learning techniques to financial risk modeling and predictive analytics. It may be particularly relevant to:

- Credit Risk Analysts
- Portfolio Risk Managers
- Data Scientists
- Quantitative Analysts
- Lending Strategy teams
- Business stakeholders who are responsible for portfolio performance

5. Data Source

This project uses the publicly available LendingClub Loan Data obtained from Kaggle. The dataset contains approximately 2.26 million loan records and 145 variables describing borrower characteristics, loan attributes, credit history, payment behavior, and loan outcomes.

The target variable was derived from the original loan status field by creating a binary classification. Loans with the statuses “Charged Off”, “Default”, and “Does Not Meet the Credit Policy: Charged Off” were labeled as Default (1), while all remaining loan statuses were labeled as Non-default (0).

The large size and rich set of financial variables make this dataset well-suited for developing and evaluating machine learning models for predicting the probability of default (PD).

Dataset:
Kaggle – LendingClub Loan Data

Due to the size of the full dataset, model development was performed on a representative random sample of 75,000 observations, preserving the original class distribution. This enabled efficient experimentation and model comparison while preserving the full dataset's representative characteristics.

6. Data Preparation & Feature Engineering

Before model development, the raw LendingClub dataset underwent several preprocessing and refinement steps to create a suitable dataset for binary classification.

6.1 Target Variable Definition

The original loan_status variable was used to construct the binary target variable. Loans classified as Charged Off, Default, or Does Not Meet the Credit Policy: Charged Off were assigned to the default class (1). All remaining loan statuses were classified as non-default (0). After the target variable was created, the original loan_status column was removed from the predictor set.

6.2 Removal of Data Leakage Variables

Variables containing information that would only become available after loan performance had already been observed were removed to prevent data leakage. These included payment, recovery, outstanding principal, and other post-origination performance variables such as total_pymnt, total_rec_prncp, total_rec_int, recoveries, collection_recovery_fee, and last_pymnt_amnt. Removing these variables ensured that the models relied on information appropriate for predicting default rather than information that directly revealed the eventual loan outcome.

6.3 Missing-Value Assessment and Column Filtering

The dataset was then evaluated for missing values. Columns with more than 20% missing values were removed, retaining variables with at least 80% of data available. This reduced the feature set from 145 original variables to 76 columns, creating a more manageable dataset while retaining variables with sufficient information for subsequent analysis.

6.4 Dataset Refinement

Following the initial missing-value filtering, the dataset was further refined by removing variables unlikely to improve model performance or that could introduce noise, bias, or data leakage. High-cardinality identifier and text-based variables, including zip_code, title, and emp_title, were excluded because they contained large numbers of unique values and were therefore not generalizable. Near-constant variables, such as pymnt_plan and policy_code, were also removed due to their limited predictive value.

Additionally, variables containing information generated after loan origination, including last_credit_pull_d and debt_settlement_flag, were excluded to prevent the model from learning from information that would not have been available at the relevant prediction point. Following this refinement, the dataset contained 69 remaining variables.

6.5 Missing-Value Treatment

Missing values were handled according to the meaning and characteristics of each variable rather than applying a single imputation method across the dataset.
For emp_length, categorical employment-duration values were converted into a numerical representation. Because missing employment history may itself contain predictive information, an additional binary indicator (emp_length_missing) was created to distinguish borrowers with missing employment length information. Missing values in the original employment-length variable were then encoded as -1.
Numerical variables were divided into three broad groups and treated accordingly:
Event-based variables: Missing values indicating that an event had not occurred were replaced with 0.
Time-since-event variables: Where missingness represented that a particular credit event had never occurred, values were encoded as -1. This allowed the models to distinguish between recent events, older events, and no recorded occurrence.
Continuous financial and account variables: Missing values were imputed using the median, which provides a robust measure of central tendency and reduces sensitivity to extreme values and outliers.
This variable-specific approach preserved meaningful information contained in missing values while producing a complete numerical dataset suitable for subsequent preprocessing and model development.

6.6 Variable Encoding and Feature Transformation

Following missing-value treatment, the remaining variables were transformed into formats suitable for machine learning. Object-type variables were individually reviewed to determine whether they represented categorical, ordinal, numerical, or date-based information, and were encoded accordingly.
Several variables were directly converted to numerical representations. The loan term variable was converted from text values such as “36 months” and “60 months” into integers. Binary categorical

variables, including initial_list_status, hardship_flag, application_type, and disbursement_method, were mapped to 0/1 indicators.

Ordinal credit-quality variables were encoded while preserving their natural ordering. Loan grades were mapped numerically from A = 1 to G = 7, and the numerical component of each subgrade was extracted and combined with its corresponding grade to create a grade_combined feature. For example, A1 was represented as 1.1 and B3 as 2.3. This provided a single numerical measure capturing the ordered structure of LendingClub's credit grades and subgrades.

Categorical variables with multiple levels were consolidated where appropriate to reduce sparsity. Rare home_ownership categories were grouped into an "Other" category, while detailed loan purposes were consolidated into broader groups such as debt, home, purchase, business, and personal. The remaining nominal categorical variables were converted using **one-hot encoding**, allowing them to be incorporated into the models without imposing an artificial numerical ordering. Boolean variables were subsequently converted to integer indicators.

6.7 Time-Based Feature Engineering

Date variables were converted from text into a datetime format and used to construct additional time-based features. The dataset contained loans issued between June 2007 and December 2018, while borrowers' earliest recorded credit lines dated back further.
A fixed reference date of January 1, 2019, was used to construct time-based features rather than using the current date. This ensured that the derived variables reflected information relative to the dataset's historical period and avoided introducing information from outside the observation window.

Two features were then constructed:
loan_age_years — the number of years between the loan's issue date and the reference date.
credit_history_years — the number of years between the borrower's earliest recorded credit line and the reference date.
These transformations converted the original date fields into interpretable numerical measures capturing loan seasoning and the length of a borrower's credit history. After creating these time-based features, the original issue_d and earliest_credit_line variables were removed, along with the addr_state variable. After preprocessing and feature engineering, all remaining variables were converted to numerical format, and the dataset contained no missing values. The resulting dataset was therefore ready for exploratory data

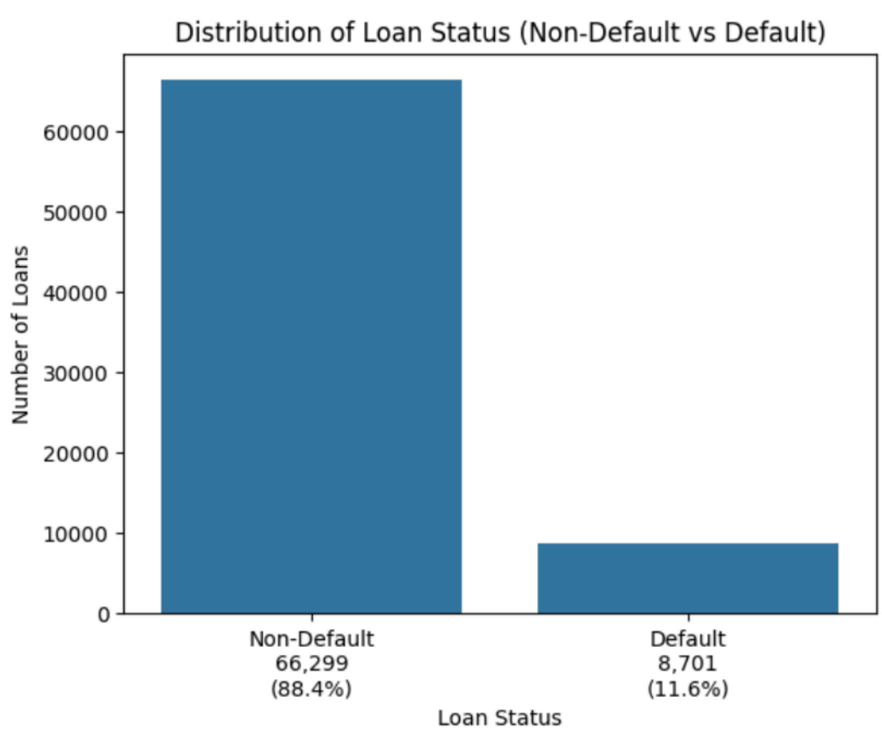
analysis and subsequent machine learning model development.

7. Exploratory Data Analysis & Key Findings

The initial stage of the analysis involved conducting exploratory data analysis (EDA) to examine associations among borrower attributes, loan characteristics, credit quality, and the likelihood of default. Utilizing a combination of correlation analysis, box plots, and empirical cumulative distribution function (ECDF) visualizations, the objective was to identify significant patterns and potential predictive signals to inform subsequent model development

7.1 Target Variable Analysis and Class Imbalance

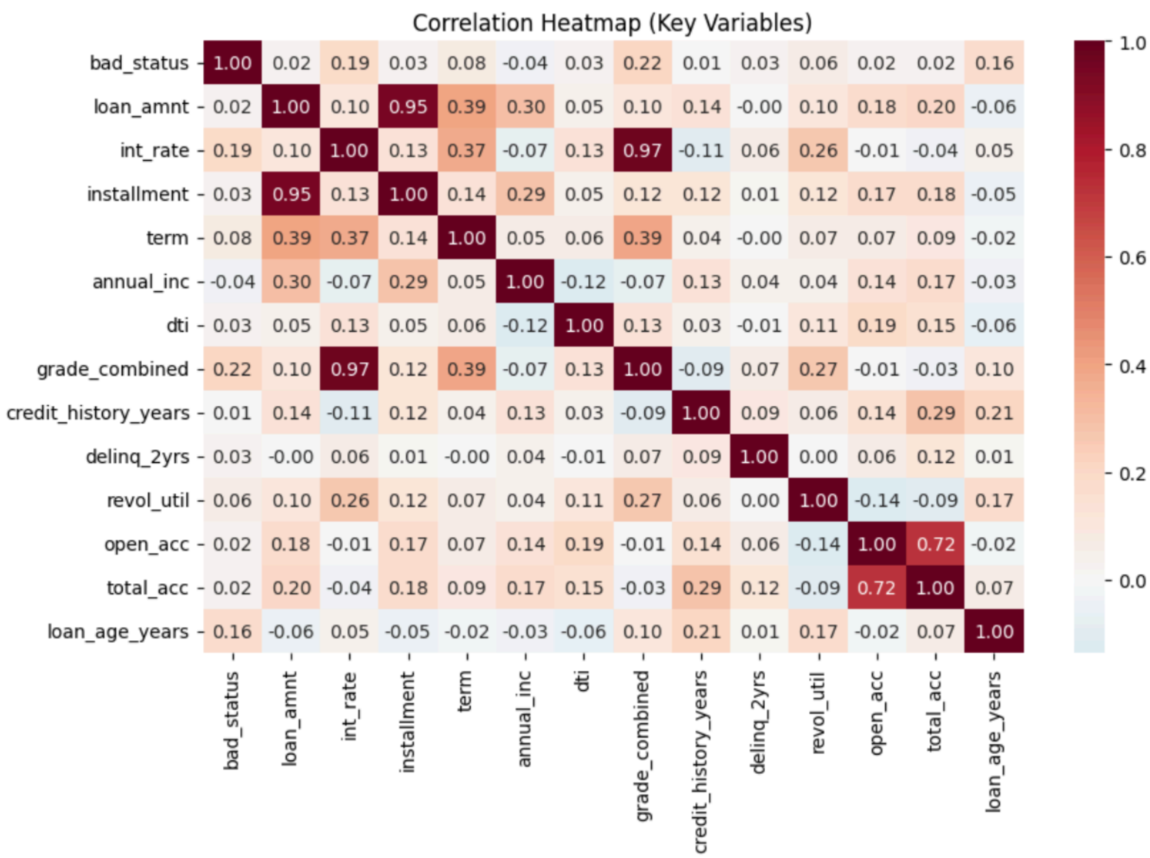
A review of the target variable revealed substantial class imbalance: approximately 88.4% of observations were classified as Non-default (0), and 11.6% as Default (1). Such a distribution presents a challenge for machine learning algorithms, as a model might achieve high overall accuracy by disproportionately favoring the majority class while failing to identify defaulted loans adequately. Consequently, the modeling process required the use of specific performance metrics and sampling techniques tailored for imbalanced datasets.



7.2 Correlation Analysis and Multicollinearity

A comprehensive correlation analysis was performed across borrower financial metrics, credit-quality indicators, and derived time-based features. The results showed that **grade_combined** and **interest_rate** had the strongest positive correlations with default, indicating that higher interest rates and lower credit grades are primary indicators of risk.

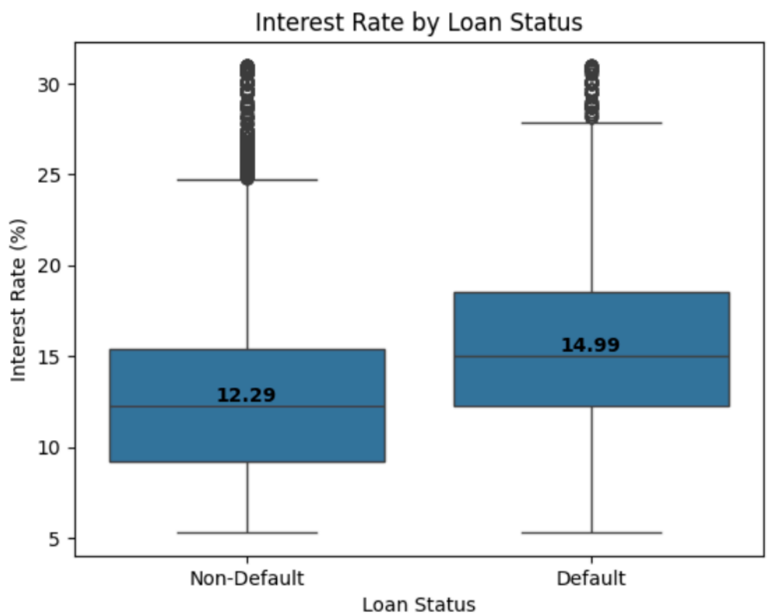
Furthermore, the analysis identified significant intercorrelations among several independent variables, notably between loan amount and monthly installments, and between credit grade and interest rate. These findings suggest the presence of multicollinearity, which must be accounted for when interpreting the coefficients of linear models such as logistic regression.



7.3 Relationship Between Interest Rates and Default

The analysis confirmed a distinct relationship between interest rates and loan outcomes. Box plots showed that defaulted loans typically have higher interest rates, as evidenced by an upward shift in the overall distribution and a higher median than non-defaulted accounts.

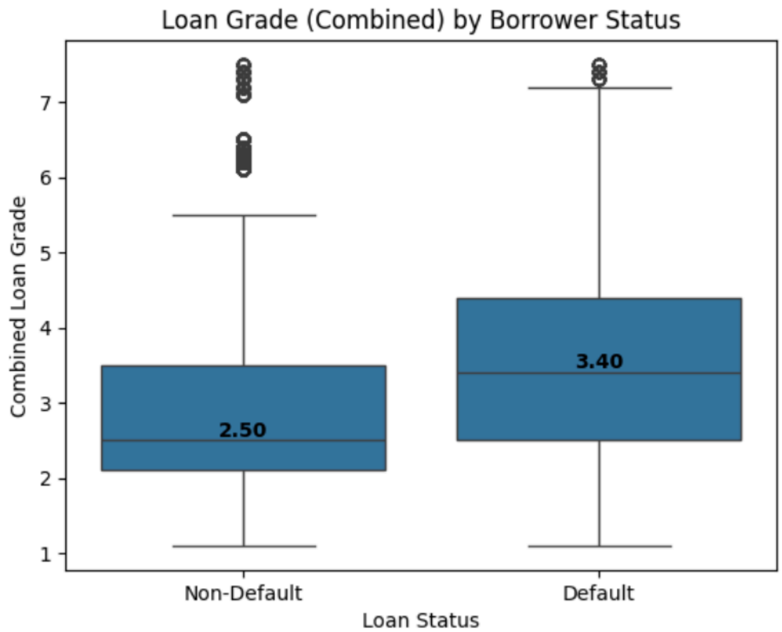
This trend was further supported by the ECDF analysis, which showed that the distribution of default interest rates is notably skewed toward higher values. These observations suggest that the interest rate assigned at origination is a highly informative indicator of the subsequent default probability.



7.4 Loan Grade and Default Risk

The combined loan grade showed a clear relationship with default status. Defaulted loans had a median combined grade of approximately 3.4 (C4), compared with approximately 2.5 (B5) for non-defaulted loans, indicating that loans with weaker credit grades were more likely to default. However, considerable overlap remained between the two groups, demonstrating that credit grade alone does not determine loan outcomes.

Combined loan grade was also highly correlated with the interest rate ($r \approx = 0.97$), indicating substantial multicollinearity between the two variables. This relationship was considered when interpreting subsequent model results, particularly Logistic Regression coefficients.



7.5 Debt-to-Income Ratio

Defaulted loans exhibited slightly higher debt-to-income (DTI) ratios than non-defaulted loans. Although the median DTI was higher among defaulted borrowers, substantial overlap between the distributions suggests that DTI alone provides limited separation between the two classes. Extreme values were present in the data; therefore, the displayed y-axis was restricted to improve visualization of the main distribution.

7.6 Annual Income

Annual income showed little separation between defaulted and non-defaulted loans, with nearly identical median values and substantial overlap between the two distributions. A logarithmic transformation was used for visualization due to the presence of extreme income values; however, differences between the two groups remained minimal. This suggests that annual income alone has a weak direct relationship with default risk in this dataset.

7.7 Loan Purpose

Loan purpose distributions were broadly similar between defaulted and non-defaulted loans. Debt-related borrowing accounted for the largest share in both groups, while the remaining purpose categories showed relatively similar proportions. Overall, loan purposes did not appear to provide strong separation between default and non-default outcomes.

Overall, the exploratory analysis identified credit grade and interest rate as the variables most clearly associated with default status. At the same time, DTI showed a weaker relationship, and annual income and loan purpose provided relatively little separation between the two classes. The analysis also revealed substantial class imbalance and strong correlation among several predictors, particularly between credit grade and interest rate. These findings informed subsequent model development, evaluation, and interpretation.

8. Model Development

8.1 Train-Test Split

The prepared dataset was split into training and test sets using an 80/20 stratified split, preserving the proportions of default and non-default loans in both subsets. Stratification was particularly important given the class imbalance in the target variable and ensured that both datasets maintained a similar class distribution.

8.2 Feature Scaling

Numerical features were standardized using StandardScaler to place variables with different magnitudes on a comparable scale. The scaler was fitted exclusively on the training data and subsequently applied to both the training and testing sets. In this way, the information from the test set won't influence preprocessing decisions during model training.

8.3 Modeling

Several classification approaches were developed and compared, including Logistic Regression, Random Forest, and XGBoost. Multiple model variants were evaluated to address class imbalance and assess the effect of model tuning. These included baseline models, class-weighted approaches, SMOTE-based resampling, and hyperparameter tuning.

Logistic Regression provided an interpretable benchmark, while Random Forest and XGBoost were evaluated to capture more complex nonlinear relationships and interactions among predictors. Hyperparameter tuning was performed on selected models to determine whether predictive performance could be improved beyond the baseline specifications.

9. Model Evaluation, Comparison, and Final Model Selection

9.1 Performance Summary

Several machine learning models and modeling approaches were developed, evaluated, and refined throughout this project to predict loan default. This section compares the best-performing versions of the evaluated models using key performance metrics to determine which approach offers the best balance between default detection, overall classification performance, and model interpretability.

Because the dataset is highly imbalanced, accuracy alone may provide a misleading assessment of model performance. A model can achieve relatively high accuracy by correctly classifying the majority non-default class while failing to identify a substantial proportion of defaulted loans. Therefore, model performance was evaluated using recall for both the default and non-default classes, F1 score, ROC-AUC, and the Gini coefficient in addition to overall accuracy. Default recall is particularly important in this analysis because it measures the model's ability to identify loans that ultimately default. In contrast, ROC-AUC and Gini measure the model's overall ability to discriminate between defaulted and non-defaulted loans.

Comparison of Best-Performing Classification Models

Model	Accuracy	Recall: Non-Default (Class 0)	Recall: Default (Class 1)	F1-Score: Default (Class 1)	ROC-AUC	Gini
Balanced Logistic Regression	70.0%	70.0%	68.0%	34.0%	75.9%	0.518
Tuned Logistic Regression	69.0%	69.0%	69.0%	34.0%	75.8%	0.516
Tuned Random Forest	66.0%	65.0%	73.0%	33.0%	75.8%	0.516
Baseline XGBoost	69.0%	69.0%	71.0%	35.0%	77.2%	0.543
Tuned XGBoost	71.0%	71.0%	67.0%	35.0%	77.0%	0.539

The table compares the five selected model specifications: Balanced Logistic Regression, Tuned Logistic Regression, Tuned Random Forest, Baseline XGBoost, and Tuned XGBoost

9.2 Performance Comparison

The performance comparison highlights the trade-offs among the five best-performing classification models. No single model achieved the highest score across every evaluation metric. Model selection, therefore, requires consideration of both the ability to identify defaulted loans and the ability to distinguish between defaulted and non-defaulted loans correctly.

Tuned Random Forest. The tuned Random Forest achieved the highest recall for the default class, approximately 73%, indicating that it identified the largest proportion of loans that ultimately defaulted. However, this improvement came at the expense of lower performance on non-defaulted loans, with a non-default recall of approximately 65% and the lowest overall accuracy among the compared models at approximately 66%. This indicates that the model classified a larger proportion of non-defaulted loans as defaults, leading to more false positives.

Balanced Logistic Regression. Balanced Logistic Regression produced relatively balanced performance across the two classes, achieving approximately 70% accuracy, 70% recall for non-defaulted loans, and 68% recall for defaulted loans, while maintaining a competitive ROC-AUC of approximately 0.759. These results demonstrate that class weighting substantially improved the model's ability to identify defaults while maintaining reasonable performance on the majority class.

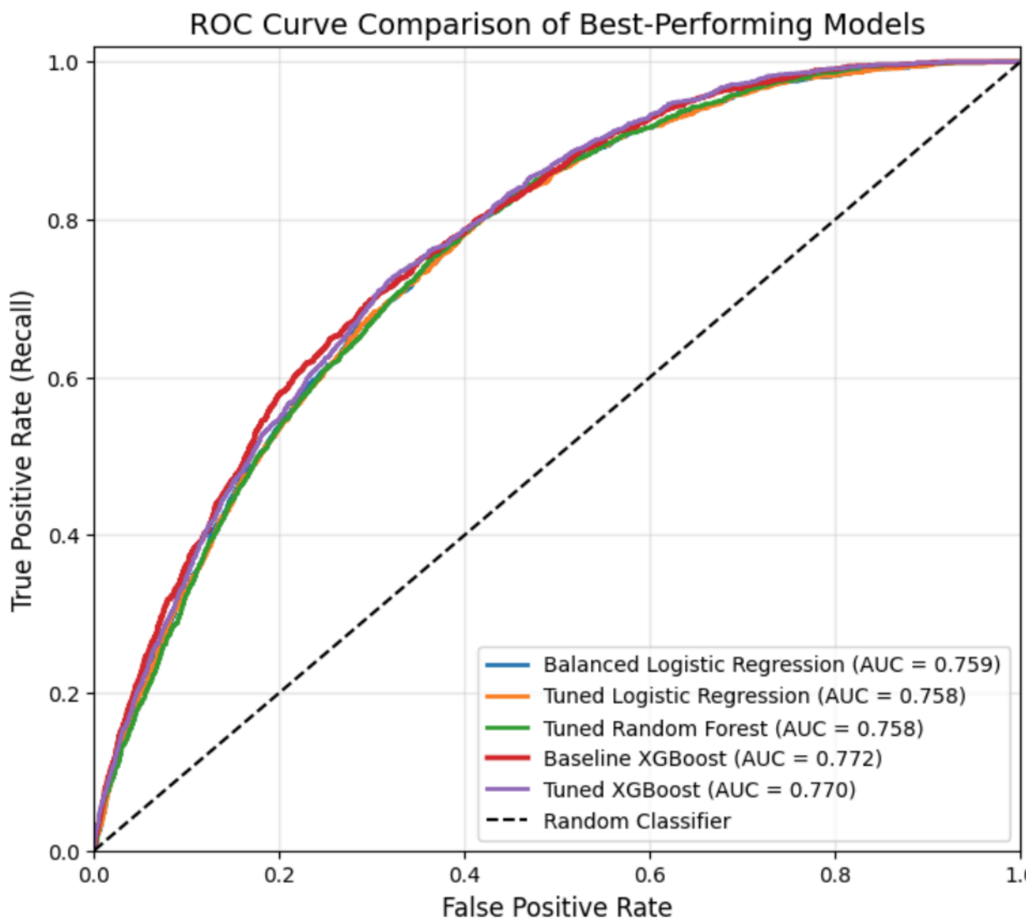
Tuned Logistic Regression. Hyperparameter-tuned Logistic Regression produced results that were very similar to Balanced Logistic Regression. The limited difference in performance suggests that additional hyperparameter tuning provided only marginal improvement for Logistic Regression on this dataset.

Baseline XGBoost. Baseline XGBoost demonstrated the strongest overall discriminatory performance among the evaluated models. It achieved the highest ROC-AUC of approximately 0.772 and a Gini coefficient of approximately 0.543, while maintaining a strong default recall of approximately 71% and an F1 score of approximately 35%. Although Tuned Random Forest achieved slightly higher default recall, Baseline XGBoost provided a stronger overall balance between identifying defaulted loans and discriminating between the two outcome classes.

Tuned XGBoost. Tuned XGBoost achieved slightly higher overall accuracy than the baseline specification, approximately 71% compared with 69%. However, default recall decreased to approximately 67%, while ROC-AUC and Gini were marginally lower at approximately 0.770 and 0.539, respectively. Given the objective of identifying loans at elevated risk of default, the increase in overall accuracy did not compensate for the reduction in default recall. The relatively small differences between the baseline and tuned XGBoost models also suggest that hyperparameter tuning did not materially improve predictive performance on this dataset.

9.3 ROC Curve Comparison

The Receiver Operating Characteristic (ROC) curve provides an additional way to compare the discriminatory ability of the classification models across different classification thresholds. The curve plots the true positive rate (recall) against the false positive rate. The diagonal reference line represents the performance of a random classifier, with an ROC-AUC of 0.50. A model whose ROC curve lies further above the diagonal and closer to the upper-left corner demonstrates a stronger ability to distinguish between defaulted and non-defaulted loans. The ROC-AUC summarizes this performance across all classification thresholds, with higher values indicating better overall discrimination.



The figure compares the ROC curves of the five best-performing classification models. All curves lie well above the diagonal reference line, indicating that each model demonstrates meaningful discriminatory ability beyond random classification. ROC-AUC values range from approximately 0.758 to 0.772, and the substantial overlap among the curves indicates relatively similar discriminatory performance across the evaluated models.

Baseline XGBoost achieved the highest ROC-AUC at approximately 0.772, although the differences between the models were relatively small. This reinforces the importance of considering additional performance measures, particularly default recall, rather than selecting the final model solely based on ROC-AUC.

10. Final Model Selection

Based on the comparative evaluation, **Baseline XGBoost was selected as the final model**. Although Tuned Random Forest achieved the highest default recall, Baseline XGBoost maintained a strong default recall of approximately 71% while achieving the highest ROC-AUC and Gini coefficient among the evaluated models. This indicates stronger overall discriminatory ability while still identifying a substantial proportion of defaulted loans.

Given the project's objective, Baseline XGBoost provided the most appropriate balance between default detection and overall model discrimination. Its performance also demonstrates why model selection should not be based on a single evaluation metric, especially when working with an imbalanced dataset.

Balanced Logistic Regression remains a useful alternative when model transparency and interpretability are prioritized; it's relatively balanced recall across both classes, and the competitive ROC-AUC shows that a simpler, more interpretable model can still achieve meaningful discrimination.

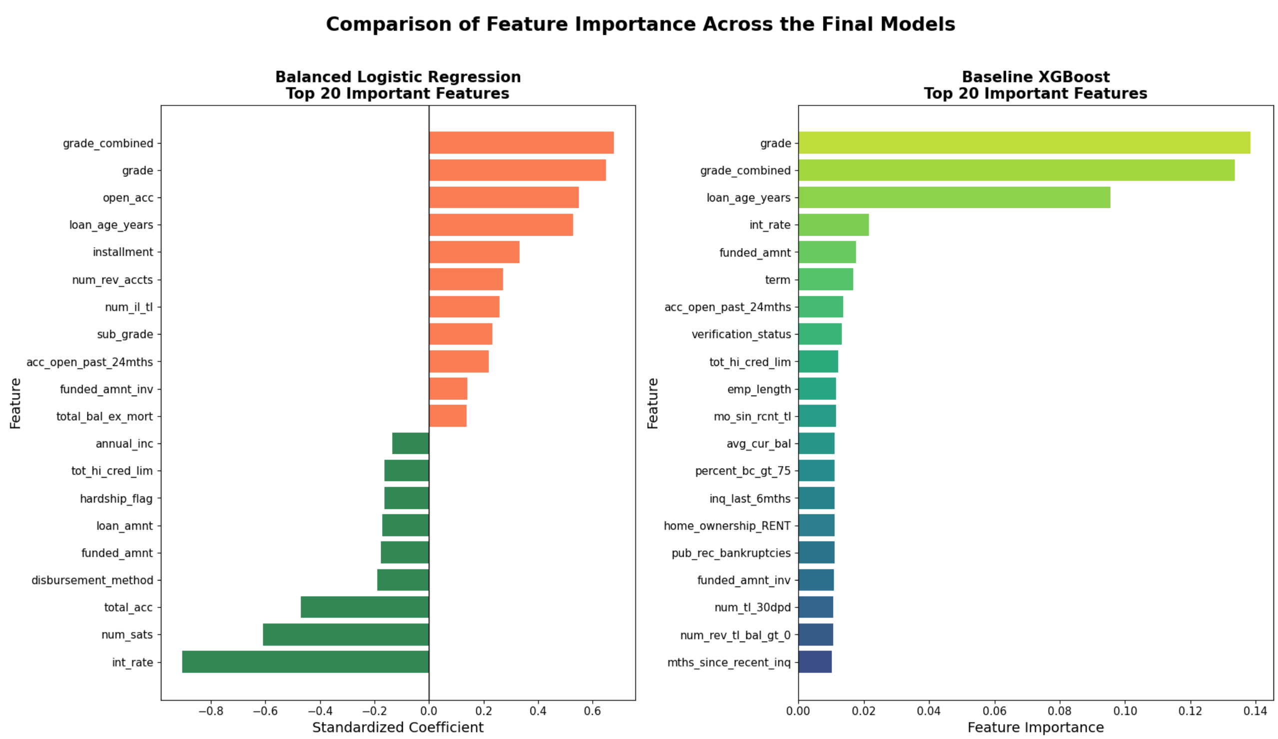
11. Feature Importance and Interpretation

To better understand the factors contributing to model predictions, feature importance was examined for the selected Baseline XGBoost model and compared with Balanced Logistic Regression. Baseline XGBoost represents the final selected model, while Balanced Logistic Regression provides a simpler and more interpretable benchmark with competitive predictive performance. Comparing the two models helps identify whether similar borrower and loan characteristics are considered important across different modeling approaches.

For the Balanced Logistic Regression model, features were ranked using the absolute values of their standardized coefficients. Absolute values were used for ranking so that features with large positive or negative coefficients could both be identified as influential. The original coefficient signs were retained for interpretation: positive coefficients indicate that higher values of a feature are associated with a higher predicted probability of default. In contrast, negative coefficients indicate an association with a lower predicted probability of default, holding the other variables constant.

The negative coefficient observed for interest rate should not be interpreted as evidence that higher interest rates reduce default risk. The interest rate is highly correlated with the combined loan grade (r ≈ 0.97), resulting in substantial multicollinearity. Because the two variables capture closely related information, their individual Logistic Regression coefficients may become unstable or change sign when included together. The negative interest-rate coefficient should therefore be interpreted cautiously.

For Baseline XGBoost, feature importance scores measure the relative contribution of individual features to the model's predictions but do not indicate whether a feature increases or decreases predicted default risk.



Key Findings

Loan grade and loan age were among the most influential predictors across both models. Loan grade captures information about a borrower's credit quality and is closely associated with the interest rate assigned to a loan, making it an important indicator of underlying credit risk.

Beyond loan grade and loan age, the models identified a range of additional borrower, loan, and credit-history characteristics as important. Balanced Logistic Regression emphasized variables such as the number of open and revolving accounts, installment amount, recently opened accounts, annual income, loan amount, and interest rate, among others. Baseline XGBoost identified some of the same predictors while also placing importance on other characteristics related to credit activity, delinquency, and public-record history, employment information, verification status, recent credit inquiries, and encoded categorical variables such as home-ownership status.

Several variables that required preprocessing or treatment of missing values also remained among the important predictors. Although the feature importance analysis does not indicate whether the missing values themselves contributed to their importance, the presence of these variables among the leading predictors supports retaining and preprocessing potentially useful information rather than automatically excluding them from the analysis.

The differences in feature rankings are expected because the two models identify predictive patterns differently. Logistic Regression provides a relatively straightforward estimate of how individual features are associated with predicted default risk while accounting for the other variables in the model. XGBoost can capture more complex patterns and feature interactions. As a result, the two models do not necessarily assign the same level of importance to each predictor.

The models also provide different forms of interpretation. Logistic Regression coefficients show both the relative strength and direction of a feature's relationship with predicted default risk. In contrast, XGBoost feature importance identifies which variables contribute most to the model's predictions, but by itself, does not indicate whether higher values of those variables increase or decrease default risk.

Overall, comparing the two models provides a broader understanding of the factors associated with loan default. Balanced Logistic Regression offers a more straightforward interpretation of individual feature relationships, while Baseline XGBoost can capture more complex predictive patterns and achieve stronger overall discriminatory performance.

12. Limitations and Future Work

While the models developed in this project demonstrated meaningful ability to distinguish between defaulted and non-defaulted loans, several limitations should be considered when interpreting the results.

First, the analysis was conducted using a sample of the full LendingClub dataset rather than the complete dataset of more than two million loans. Although the sample allowed for efficient model development and comparison, training and evaluating the final model on the full dataset would help assess scalability and determine whether the observed performance generalizes to a substantially larger sample.

Second, the models were evaluated using a stratified train-test split that preserved the proportion of defaulted and non-defaulted loans. While this provides a consistent evaluation framework, it does not assess how well the model performs on loans issued later. Credit risk relationships may change as borrower behavior, lending practices, and economic conditions evolve. Future work should therefore include out-of-time validation using loans issued in different years.

Third, the models primarily relied on the available borrower, loan, and credit-history characteristics after data cleaning and preprocessing. Additional feature engineering, including financial ratios and interaction variables, could more effectively capture borrower risk.

Future work could also evaluate additional gradient-boosting algorithms such as LightGBM and CatBoost, which may provide competitive predictive performance and greater computational efficiency on large datasets.

The classification threshold represents another area for further development. The models in the final comparison used the default classification threshold rather than a threshold optimized for a specific business objective. In a practical credit-risk setting, the threshold could be adjusted to reflect the relative costs of failing to identify a default versus incorrectly flagging a non-defaulted loan as high risk.

Finally, additional model-explainability techniques, such as SHAP values, could be incorporated to provide more detailed insight into how individual features influence specific loan predictions. This would complement the global feature-importance analysis and improve the interpretability of the XGBoost model.

13. Conclusion

The objective of this project was to develop and evaluate machine learning models for identifying loans at risk of default using historical LendingClub loan data. The analysis presented a challenging, imbalanced classification problem, with approximately 88% of observations belonging to the non-default class. As a result, model evaluation focused not only on overall accuracy but also on the ability to identify defaulted loans and discriminate between the two outcome classes.

Multiple modeling approaches were evaluated, including Logistic Regression, Random Forest, and XGBoost, together with class weighting, SMOTE oversampling, regularization, and hyperparameter tuning. The initial Logistic Regression model demonstrated why accuracy alone is insufficient for imbalanced classification: despite high overall accuracy, it identified very few defaulted loans. Class weighting substantially improved default detection, while SMOTE and additional Logistic Regression tuning provided relatively limited improvement.

Random Forest demonstrated that maximizing a single performance metric does not necessarily produce the strongest overall model. The Tuned Random Forest achieved the highest default recall among the evaluated models, identifying approximately 73% of defaulted loans. However, this came with lower non-default recall, lower overall accuracy, and weaker discriminatory performance than the selected XGBoost model.

Baseline XGBoost ultimately provided the strongest overall balance across the evaluation criteria. It achieved the highest ROC-AUC and Gini coefficient while maintaining a strong default recall of approximately 71%. Although Tuned XGBoost achieved slightly higher overall accuracy, its default recall decreased, and its discriminatory performance did not improve. Baseline XGBoost was therefore selected as the final model.

Feature-importance analysis further showed that loan grade and loan age were among the most influential predictors across both Baseline XGBoost and Balanced Logistic Regression, while additional borrower, loan, and credit-history characteristics also contributed to model predictions. The comparison with Logistic Regression provided additional interpretability and demonstrated that different modeling approaches can identify both common and distinct patterns associated with loan default.

Overall, the project demonstrates the importance of evaluating credit risk models using multiple performance measures rather than relying on accuracy alone. The selected Baseline XGBoost model provides a useful framework for identifying loans with elevated default risk, while further validation, threshold optimization, and model explainability would be necessary before considering application in a real-world credit-risk setting.